

Divide Mixed Numbers

Lesson 2-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$2\frac{1}{3}$ means 2 wholes and $\frac{1}{3}$ more — picture 2 full circles and $\frac{1}{3}$ of another

Mixed Number

A whole number plus a fraction, like $2\frac{1}{2}$.

$\frac{7}{3}$ is improper because $7 > 3$. It equals $2\frac{1}{3}$ as a mixed number.

Improper Fraction

A fraction where the top is bigger than or equal to the bottom, like $\frac{7}{4}$.

$2\frac{1}{3} \rightarrow$ multiply $2 \times 3 = 6$, add 1 $\rightarrow \frac{7}{3}$. Same value, different form.

Convert

To change a number to a new form but keep the same value.

$\frac{8}{6}$: GCF is 2 $\rightarrow \frac{8 \div 2}{6 \div 2} = \frac{4}{3} = 1\frac{1}{3}$

Simplify

To make a fraction smaller using the same parts, like $\frac{2}{4} = \frac{1}{2}$.

The reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$. To divide by $\frac{3}{4}$, multiply by $\frac{4}{3}$.

Reciprocal

A fraction turned upside down. You use it in keep-change-flip.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can divide mixed numbers by first changing them to improper fractions.

1 Notice

Agent Chen is mapping a suspect's route.

2 Key idea

I can divide mixed numbers by first changing them to improper fractions.

3 Words I'll use

Mixed Number

Improper Fraction

Convert

Simplify

Reciprocal



Think About It

- What is the total distance of the route?
- How long is each segment?
- Will the number of segments be more or less than 3?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: What is $1\frac{1}{2} \div \frac{3}{4}$?

- A. 2
- B. $1\frac{1}{8}$
- C. $\frac{3}{4}$
- D. $\frac{9}{8}$

Step 1 $1\frac{1}{2} = \frac{3}{2}$.

Step 2 Then $\frac{3}{2} \div \frac{3}{4} = \frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = 2$.



Answer: A. 2

 We do — together

Solve this one with your class using the same steps.

Problem: Convert $3\frac{2}{5}$ to an improper fraction.

- A. $\frac{17}{5}$
- B. $\frac{15}{5}$
- C. $\frac{6}{5}$
- D. $\frac{32}{5}$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: What is $2\frac{1}{4} \div \frac{3}{4}$?

- A. 3
- B. $1\frac{1}{2}$
- C. $\frac{9}{16}$
- D. $2\frac{1}{3}$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Agent Chen maps a $3\frac{1}{2}$ -mile route divided into $\frac{1}{4}$ -mile segments. Why can't she just divide $3\frac{1}{2}$ by $\frac{1}{4}$ while it is still a mixed number?

Level 1 support

Start here: A mixed number must first become an improper fraction so I can flip and multiply.

 **Need a hint?**

Sentence starters (optional)

I cannot divide yet because $3\frac{1}{2}$ is a ____ and I need an ____.

No puedo dividir todavía porque $3\frac{1}{2}$ es un ____ y necesito una ____.

First I must convert $3\frac{1}{2}$ into ____.

Primero debo convertir $3\frac{1}{2}$ en ____.

Word bank: mixed number improper fraction convert

Listen for: Students explain that the mixed number must be converted to an improper fraction ($\frac{7}{2}$) before applying Keep, Change, Flip.

Level 2

Estimate the number of segments first. Is your estimate close to 14? Justify using the route length and segment size.

- I estimate about ____ segments because ____.
- Each mile holds ____ quarter-mile segments, so ____.

EXPLORE

Walk your partner through how you changed the mixed numbers before dividing.

Level 1 support

Start here: I converted each mixed number to an improper fraction, then used Keep, Change, Flip.



Need a hint?

Sentence starters (optional)

First I converted ____ into the improper fraction ____.

Primero convertí ____ en la fracción impropia ____.

Then I flipped the divisor and got the quotient ____.

Luego volteé el divisor y obtuve el cociente ____.

Word bank: **mixed number** **improper fraction** **reciprocal** **simplify**

Listen for: Students correctly convert (multiply whole by denominator, add numerator) before flipping the divisor and simplifying.

Level 2

A student converted $2\frac{1}{3}$ to $\frac{6}{3}$ instead of $\frac{7}{3}$. Critique exactly what step they skipped.

- The student forgot to ____.
- The correct improper fraction is ____ because ____.

CONNECT

When would dividing mixed numbers come up in a real project, like building or cooking?

Level 1 support

Start here: Whenever you split a mixed-number amount into fraction-size pieces.

 **Need a hint?**

Sentence starters (optional)

I would divide mixed numbers when ____.

Dividiría números mixtos cuando ____.

For example, cutting ____ feet into ____-foot pieces gives ____.

Por ejemplo, cortar ____ pies en piezas de ____ pie da ____.

Word bank: **mixed number** **improper fraction** **convert**

Listen for: Students give a realistic project scenario and recognize the mixed-number amount must be converted first.

Level 2

Generalize the full set of steps for dividing any two mixed numbers, in order.

- My steps are: first ____, then ____, then ____.
- This always works because ____.

Write About the Math

The Writing Revolution

I can explain my steps using the words mixed number, improper fraction, convert, reciprocal, and simplify.

1. Kernel Sentence subject + verb

Model: Mixed Number is a whole number plus a fraction, like $2\frac{1}{2}$.

Número mixto es un número entero más una fracción, como $2\frac{1}{2}$.

Write a kernel sentence about mixed number. Use a subject and a verb.

Escribe una oración base sobre número mixto. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Mixed Number matters in math

Número mixto importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Mixed Number matters in math because ____.

Número mixto importa en matemáticas porque ____.

but
pero

Mixed Number matters in math, but ____.

Número mixto importa en matemáticas, pero ____.

so
entonces

Mixed Number matters in math, so ____.

Número mixto importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about mixed number.
Di un hecho verdadero sobre mixed number.

Mixed number ____.

Question
Pregunta

Ask a question about mixed number.
Haz una pregunta sobre mixed number.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about mixed number.
Muestra entusiasmo sobre mixed number.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with mixed number.
Dile a un compañero qué hacer con mixed number.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I cannot divide yet because $3\frac{1}{2}$ is a ____ and I need an ____.

No puedo dividir todavía porque $3\frac{1}{2}$ es un ____ y necesito una ____.

First I must convert $3\frac{1}{2}$ into ____.

Primero debo convertir $3\frac{1}{2}$ en ____.

First I converted ____ into the improper fraction ____.

Primero convertí ____ en la fracción impropia ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A recipe needs $1\frac{1}{2}$ cups of flour. How many full recipes can you make with 6 cups?

- A. 4
- B. 3
- C. 5
- D. 9

Show your work:

2. What is $2\frac{1}{3} \div \frac{2}{3}$?

- A. $3\frac{1}{2}$
- B. $1\frac{2}{3}$
- C. 2
- D. 3

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective has a rope that is $5 \frac{1}{4}$ feet long. She needs to cut it into pieces that are each $\frac{3}{4}$ foot for tying evidence tags. How many pieces can she cut? Write the equation, show each step (convert, keep-change-flip, multiply, simplify), and verify your answer.

Sentence starter: First I convert $5 \frac{1}{4}$ to _____. Then I use KCF: _____ $\times$ _____ = _____. The answer is _____ pieces. I can verify because _____ $\times \frac{3}{4} = 5 \frac{1}{4}$.

Show your work:

Reflect — Exit Ticket

What is $2 \frac{2}{3} \div \frac{2}{3}$?

- A. 4
- B. $\frac{2}{3}$
- C. $\frac{8}{3}$
- D. $1 \frac{7}{9}$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $17/5 - 3 \times 5 = 15$, then $15 + 2 = 17$. So $3 \frac{2}{5} = 17/5$.
2. **You Try (notes):** A. $3 - 2 \frac{1}{4} = 9/4$. Then $9/4 \div 3/4 = 9/4 \times 4/3 = 36/12 = 3$.
3. **Try It 1:** A. $4 - 6 \div (3/2) = 6 \times (2/3) = 12/3 = 4$ recipes.
4. **Try It 2:** A. $3 \frac{1}{2} - 7/3 \div 2/3 = 7/3 \times 3/2 = 21/6 = 7/2 = 3 \frac{1}{2}$.
5. **Exit Ticket:** A. $4 - 2 \frac{2}{3} = 8/3$. Then $8/3 \div 2/3 = 8/3 \times 3/2 = 24/6 = 4$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Mixed Number is a whole number plus a fraction, like $2 \frac{1}{2}$.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).